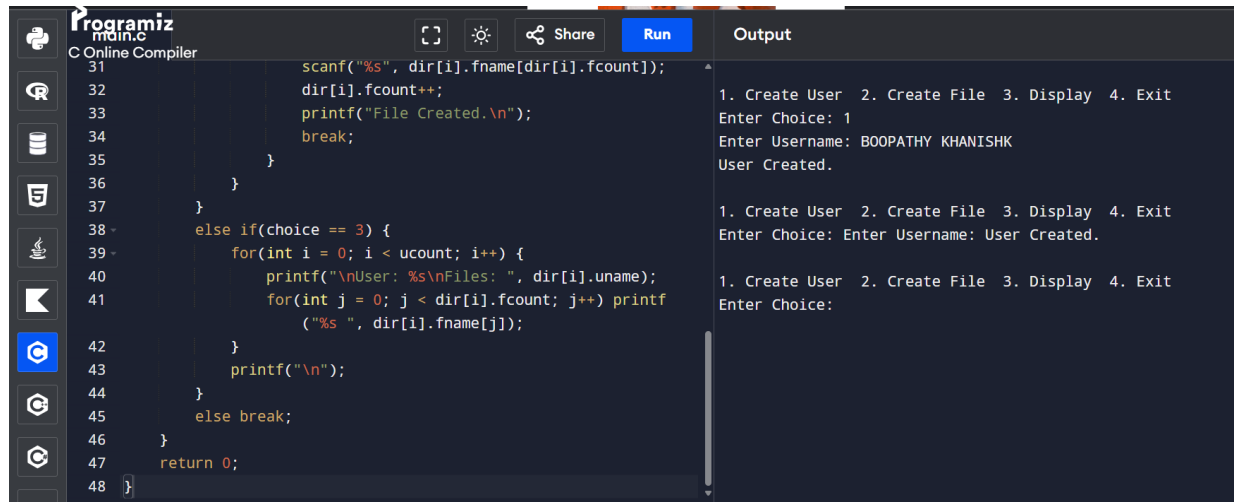


EXPERIMENT 15:

15. Design a C program to organize the file using two level directory structure.



The screenshot shows a C program in an online compiler. The code defines a structure for a directory entry with fields for username, filename, and file count. It implements a menu-driven interface with options to create users, create files, display the directory structure, and exit. The output shows the program running successfully, creating a user and a file, and displaying the directory structure.

```
31 scanf("%s", dir[i].fname[dir[i].fcount]);
32 dir[i].fcount++;
33 printf("File Created.\n");
34 break;
35 }
36 }
37 }
38 else if(choice == 3) {
39     for(int i = 0; i < ucount; i++) {
40         printf("\nUser: %s\nFiles: ", dir[i].uname);
41         for(int j = 0; j < dir[i].fcount; j++) printf
42             ("%s ", dir[i].fname[j]);
43         printf("\n");
44     }
45     else break;
46 }
47 return 0;
48 }
```

Output:

```
1. Create User  2. Create File  3. Display  4. Exit
Enter Choice: 1
Enter Username: BOOPATHY KHANISHK
User Created.

1. Create User  2. Create File  3. Display  4. Exit
Enter Choice: Enter Username: User Created.

1. Create User  2. Create File  3. Display  4. Exit
Enter Choice:
```

PSEUDO CODE :

1. Create a Master File Directory (MFD) containing user directory structures.
2. Allow adding users to the MFD.
3. Within each user's directory, allow standard file actions (Create, Delete, Display).
4. Uniquely identify files using 'Username/Filename'.

CODE:

```
#include <stdio.h>

#include <string.h>

struct UserDirectory {

    char uname[10];

    char fname[10][10];

    int fcount;

} dir[10];

int main() {

    int ucount = 0, choice;

    char u[10], f[10];

    while(1) {

        printf("\n1. Create User  2. Create File  3. Display  4. Exit\nEnter Choice: ");

        scanf("%d", &choice);

        if(choice == 1) {

            printf("Enter Username: ");

            scanf("%s", dir[ucount].uname);

            dir[ucount].fcount = 0;

            ucount++;

            printf("User Created.\n");

        }

        else if(choice == 2) {

            printf("Enter Username: ");
```

```
scanf("%s", u);

for(int i = 0; i < ucount; i++) {

    if(strcmp(u, dir[i].uname) == 0) {

        printf("Enter File Name: ");

        scanf("%s", dir[i].fname[dir[i].fcount]);

        dir[i].fcount++;

        printf("File Created.\n");

        break;

    }

}

else if(choice == 3) {

    for(int i = 0; i < ucount; i++) {

        printf("\nUser: %s\nFiles: ", dir[i].uname);

        for(int j = 0; j < dir[i].fcount; j++) printf("%s ", dir[i].fname[j]);

    }

    printf("\n");

}

else break;

}

return 0;

}
```